問題 255

次の三角形において，α の三角比を求めよ．

解答:

$\sin = \dfrac{\text{対辺}}{\text{斜辺}}$，$\cos = \dfrac{\text{隣辺}}{\text{斜辺}}$，$\tan = \dfrac{\text{対辺}}{\text{隣辺}}$

(1) 三辺 $2, 1, \sqrt{3}$（斜辺 2）

$\sin\alpha = \dfrac{1}{2}$，$\cos\alpha = \dfrac{\sqrt{3}}{2}$，$\tan\alpha = \dfrac{1}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}$

(2) 三辺 $\sqrt{7}, \sqrt{3}, 2$（$(\sqrt{3})^2 + 2^2 = 7$ より斜辺 $\sqrt{7}$）

$\sin\alpha = \dfrac{2}{\sqrt{7}} = \dfrac{2\sqrt{7}}{7}$，$\cos\alpha = \dfrac{\sqrt{3}}{\sqrt{7}} = \dfrac{\sqrt{21}}{7}$，$\tan\alpha = \dfrac{2}{\sqrt{3}} = \dfrac{2\sqrt{3}}{3}$

(3) 三辺 $12, 13, 5$（$5^2 + 12^2 = 169 = 13^2$ より斜辺 13）

$\sin\alpha = \dfrac{12}{13}$，$\cos\alpha = \dfrac{5}{13}$，$\tan\alpha = \dfrac{12}{5}$

問題 256

次の式の値を求めよ．

解答:

加法定理を逆に使って 1 つにまとめる．

(1) $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ = \sin(60^\circ - 30^\circ) = \sin 30^\circ = \dfrac{1}{2}$

(2) $\cos 30^\circ \cos 60^\circ + \sin 30^\circ \sin 60^\circ = \cos(30^\circ - 60^\circ) = \cos 30^\circ = \dfrac{\sqrt{3}}{2}$

(3) $\dfrac{\tan 60^\circ - \tan 45^\circ}{1 + \tan 60^\circ \tan 45^\circ} = \tan(60^\circ - 45^\circ) = \tan 15^\circ$
$= \dfrac{\sqrt{3} - 1}{1 + \sqrt{3}} = \dfrac{(\sqrt{3} - 1)^2}{2} = 2 - \sqrt{3}$

問題 257

三角関数表を用いて，次の三角比を求めよ．

解答:

(1) $\sin 6^\circ \approx 0.1045$　(2) $\cos 33^\circ \approx 0.8387$　(3) $\tan 84^\circ \approx 9.5144$

問題 258

高さ 634 m のスカイツリーを見上げる角度が 22° のとき，距離を求めよ．

解答:

$\tan 22^\circ = \dfrac{634}{d}$，$d = \dfrac{634}{\tan 22^\circ} \approx \dfrac{634}{0.4040} \approx 1569\,\mathrm{m}$

問題 259

次の三角比を 45° より小さい角の三角比で表せ．

解答:

(1) $\sin 81^\circ = \cos 9^\circ$　(2) $\cos 56^\circ = \sin 34^\circ$　(3) $\tan 77^\circ = \dfrac{1}{\tan 13^\circ}$

問題 260

次の式の値を求めよ．

解答:

(1) $\sin 45^\circ \cos 135^\circ - \cos 45^\circ \sin 135^\circ = \sin(45^\circ - 135^\circ) = \sin(-90^\circ) = -1$

(2) $\dfrac{\tan 45^\circ - \tan 150^\circ}{1 + \tan 45^\circ \tan 150^\circ} = \tan(45^\circ - 150^\circ)$

$\tan 150^\circ = -\dfrac{1}{\sqrt{3}}$ より $= \dfrac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \dfrac{\sqrt{3} + 1}{\sqrt{3} - 1} = 2 + \sqrt{3}$

(3) $\cos 120^\circ \cos 150^\circ + \tan 120^\circ \sin 150^\circ + \sin 120^\circ \tan 135^\circ$
$= \left(-\dfrac{1}{2}\right)\left(-\dfrac{\sqrt{3}}{2}\right) + (-\sqrt{3}) \cdot \dfrac{1}{2} + \dfrac{\sqrt{3}}{2} \cdot (-1) = \dfrac{\sqrt{3}}{4} - \dfrac{\sqrt{3}}{2} - \dfrac{\sqrt{3}}{2} = -\dfrac{3\sqrt{3}}{4}$

問題 261

三角関数表を用いて，次の三角比を求めよ．

解答:

$\sin(180^\circ - \theta) = \sin\theta$，$\cos(180^\circ - \theta) = -\cos\theta$，$\tan(180^\circ - \theta) = -\tan\theta$

(1) $\sin 100^\circ = \sin 80^\circ \approx 0.9848$

(2) $\cos 176^\circ = -\cos 4^\circ \approx -0.9976$

(3) $\tan 111^\circ = -\tan 69^\circ \approx -2.6051$

問題 262

次の問いに答えよ．

解答:

$\sin^2\alpha + \cos^2\alpha = 1$ で残りを求める．

(1) α 鋭角，$\sin\alpha = \dfrac{1}{4}$：$\cos\alpha = \dfrac{\sqrt{15}}{4}$，$\tan\alpha = \dfrac{1}{\sqrt{15}} = \dfrac{\sqrt{15}}{15}$

(2) α 鈍角，$\sin\alpha = \dfrac{1}{4}$：$\cos\alpha = -\dfrac{\sqrt{15}}{4}$，$\tan\alpha = -\dfrac{\sqrt{15}}{15}$

(3) α 鈍角，$\cos\alpha = -\dfrac{5}{6}$：$\sin\alpha = \dfrac{\sqrt{11}}{6}$，$\tan\alpha = -\dfrac{\sqrt{11}}{5}$

問題 263

$\tan\alpha$ から $\sin\alpha$，$\cos\alpha$ を求めよ．

解答:

$1+\tan^2\alpha=\dfrac{1}{\cos^2\alpha}$ を使う．

(1) α 鋭角，$\tan\alpha=\dfrac{1}{3}$

$\cos^2\alpha=\dfrac{9}{10}$，$\cos\alpha=\dfrac{3\sqrt{10}}{10}$，$\sin\alpha=\dfrac{\sqrt{10}}{10}$

(2) α 鈍角，$\tan\alpha=-2$

$\cos^2\alpha=\dfrac{1}{5}$，$\cos\alpha=-\dfrac{\sqrt{5}}{5}$，$\sin\alpha=\dfrac{2\sqrt{5}}{5}$

問題 264

△ABC において，正弦定理を用いて求めよ．

解答:

$\dfrac{a}{\sin A}=\dfrac{b}{\sin B}=\dfrac{c}{\sin C}=2R$

(1) $b=4$，$A=30°$，$B=45°$

$\dfrac{a}{\sin 30°}=\dfrac{4}{\sin 45°}$，$a=\dfrac{4\sin 30°}{\sin 45°}=\dfrac{2}{\frac{\sqrt{2}}{2}}=2\sqrt{2}$

(2) $b=2$，$c=\sqrt{3}$，$B=45°$

$\dfrac{2}{\sin 45°}=\dfrac{\sqrt{3}}{\sin C}$，$\sin C=\dfrac{\sqrt{3}\cdot\frac{\sqrt{2}}{2}}{2}=\dfrac{\sqrt{6}}{4}$

(3) $c=5$，$A=45°$，$B=105°$

$C=30°$. $\dfrac{a}{\sin 45°}=\dfrac{5}{\sin 30°}=10$, $a=10\cdot\dfrac{\sqrt{2}}{2}=5\sqrt{2}$

問題 265

1 辺が a の正三角形の外接円の半径を求めよ．

解答:

$2R=\dfrac{a}{\sin 60°}=\dfrac{2a}{\sqrt{3}}\to R=\dfrac{\sqrt{3}\,a}{3}$

問題 266

余弦定理を用いて辺の長さを求めよ．

解答:

$a^2=b^2+c^2-2bc\cos A$

(1) $b=4$，$c=\sqrt{3}$，$A=30°$

$a^2=16+3-8\sqrt{3}\cdot\dfrac{\sqrt{3}}{2}=19-12=7$，$a=\sqrt{7}$

(2) $a=\sqrt{3}$，$c=\sqrt{6}$，$B=135°$

$b^2=3+6-2\sqrt{18}\cdot\left(-\dfrac{\sqrt{2}}{2}\right)=9+6=15$，$b=\sqrt{15}$

(3) $a=3\sqrt{3}$，$b=2$，$C=150°$

$c^2=27+4-12\sqrt{3}\cdot\left(-\dfrac{\sqrt{3}}{2}\right)=31+18=49$，$c=7$

問題 267

$a=2$，$b=4$，$c=5$ のとき $\cos A$，$\cos B$，$\cos C$ を求めよ．

解答:

$\cos A=\dfrac{b^2+c^2-a^2}{2bc}=\dfrac{16+25-4}{40}=\dfrac{37}{40}$

$\cos B=\dfrac{a^2+c^2-b^2}{2ac}=\dfrac{4+25-16}{20}=\dfrac{13}{20}$

$\cos C=\dfrac{a^2+b^2-c^2}{2ab}=\dfrac{4+16-25}{16}=-\dfrac{5}{16}$

問題 268

△ABC の面積を求めよ．

解答:

$S=\dfrac{1}{2}bc\sin A$

(1) $b=5$，$c=7$，$A=45°$：$S=\dfrac{35}{2}\cdot\dfrac{\sqrt{2}}{2}=\dfrac{35\sqrt{2}}{4}$

(2) $a=2$，$b=3$，$C=150°$：$S=\dfrac{1}{2}\cdot 2\cdot 3\cdot\dfrac{1}{2}=\dfrac{3}{2}$

問題 269

$a=7$，$B=30°$，面積 9 のとき c を求めよ．

解答:

$9=\dfrac{1}{2}\cdot 7\cdot c\cdot\sin 30°=\dfrac{7c}{4}\to c=\dfrac{36}{7}$

問題 270

$a=5$，$b=6$，$c=9$ のとき．

解答:

(1) $\cos C=\dfrac{25+36-81}{60}=-\dfrac{1}{3}$

(2) $\sin C=\sqrt{1-\dfrac{1}{9}}=\dfrac{2\sqrt{2}}{3}$

(3) $S=\dfrac{1}{2}\cdot 5\cdot 6\cdot\dfrac{2\sqrt{2}}{3}=10\sqrt{2}$

問題 271

ヘロンの公式で面積を求めよ．

解答:

$S=\sqrt{s(s-a)(s-b)(s-c)}$，$s=\dfrac{a+b+c}{2}$

(1) $a=5$，$b=7$，$c=8$：$s=10$

$S=\sqrt{10\cdot 5\cdot 3\cdot 2}=\sqrt{300}=10\sqrt{3}$

(2) $a=2$，$b=3$，$c=4$：$s=\dfrac{9}{2}$

$S=\sqrt{\dfrac{9}{2}\cdot\dfrac{5}{2}\cdot\dfrac{3}{2}\cdot\dfrac{1}{2}}=\dfrac{3\sqrt{15}}{4}$

問題 272

次の三角形で α の三角比を求めよ.

解答:

(1) 三辺 $\sqrt{2}, \sqrt{7}, 3$（斜辺 3）

$\sin\alpha = \dfrac{\sqrt{2}}{3}$，$\cos\alpha = \dfrac{\sqrt{7}}{3}$，$\tan\alpha = \dfrac{\sqrt{2}}{\sqrt{7}} = \dfrac{\sqrt{14}}{7}$

(2) 三辺 $1, 2, \sqrt{5}$（斜辺 $\sqrt{5}$）

$\sin\alpha = \dfrac{2}{\sqrt{5}} = \dfrac{2\sqrt{5}}{5}$，$\cos\alpha = \dfrac{1}{\sqrt{5}} = \dfrac{\sqrt{5}}{5}$，$\tan\alpha = 2$

問題 273

図において，$\tan 15^\circ$ の値を求めよ.

解答:

加法定理：$\tan 15^\circ = \tan(45^\circ - 30^\circ) = \dfrac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} = \dfrac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \dfrac{\sqrt{3} - 1}{\sqrt{3} + 1} = 2 - \sqrt{3}$

問題 274

傾斜角 34°，距離 815 m のケーブルカーの標高差.

解答:

$815 \sin 34^\circ = 815 \times 0.5592 \approx 456\,\mathrm{m}$

問題 275

次の式の値を求めよ.

解答:

(1) $\sin 60^\circ \cos 30^\circ + \cos 120^\circ \sin 150^\circ + \sin 135^\circ \cos 180^\circ$
$= \dfrac{3}{4} - \dfrac{1}{4} - \dfrac{\sqrt{2}}{2} = \dfrac{1 - \sqrt{2}}{2}$

(2) $\dfrac{\tan 30^\circ - \tan 135^\circ - \tan 180^\circ}{1 + \tan 120^\circ \tan 45^\circ}$
$= \dfrac{\frac{1}{\sqrt{3}} + 1 - 0}{1 - \sqrt{3}} = \dfrac{\frac{\sqrt{3}+1}{\sqrt{3}}}{1 - \sqrt{3}} = \dfrac{\sqrt{3} + 1}{\sqrt{3}(1 - \sqrt{3})} = -\dfrac{2\sqrt{3} + 3}{3}$

問題 276

α 鈍角，$\cos\alpha = -\dfrac{2}{5}$ のとき.

解答:

$\sin\alpha = \dfrac{\sqrt{21}}{5}$，$\tan\alpha = -\dfrac{\sqrt{21}}{2}$

問題 277

α 鈍角，$\tan\alpha = -3$ のとき.

解答:

$\cos\alpha = -\dfrac{\sqrt{10}}{10}$，$\sin\alpha = \dfrac{3\sqrt{10}}{10}$

問題 278

△ABC の計算.

解答:

(1) $a = \sqrt{6}$，$B = 105^\circ$，$C = 30^\circ \to A = 45^\circ$

$2R = \dfrac{\sqrt{6}}{\sin 45^\circ} = 2\sqrt{3}$，$R = \sqrt{3}$，$c = 2\sqrt{3}\sin 30^\circ = \sqrt{3}$

(2) $a = 3$，$b = 5$，$C = 120^\circ$

$c^2 = 9 + 25 + 15 = 49$，$c = 7$，$S = \dfrac{15\sqrt{3}}{4}$

(3) $a = 3$，$b = 8$，$c = 7$

$\cos B = \dfrac{9 + 49 - 64}{42} = -\dfrac{1}{7}$，$\sin B = \dfrac{4\sqrt{3}}{7}$，$S = 6\sqrt{3}$

問題 285

次の角を表す動径を図示せよ.

解答:

360° で割った余りで位置を判定.

(1) $630^\circ = 360^\circ + 270^\circ \to 270^\circ$ の位置（y 軸負方向）

(2) $-310^\circ = 50^\circ$（第 1 象限）

(3) $490^\circ = 130^\circ$（第 2 象限）

(4) $-1120^\circ = -1120 + 4 \times 360 = 320^\circ$（第 4 象限）

(5) $2000^\circ = 2000 - 5 \times 360 = 200^\circ$（第 3 象限）

問題 286

次の角は第何象限の角か.

解答:

(1) 210°：第 3　(2) 400° = 40°：第 1　(3) −740° = 340°：第 4

(4) 820° = 100°：第 2　(5) −635° = 85°：第 1

問題 287

次の三角関数の値を求めよ.

解答:

(1) $\sin 210^\circ = -\sin 30^\circ = -\dfrac{1}{2}$

(2) $\cos 570^\circ = \cos 210^\circ = -\dfrac{\sqrt{3}}{2}$

(3) $\tan 390^\circ = \tan 30^\circ = \dfrac{\sqrt{3}}{3}$

(4) $\sin 630^\circ = \sin 270^\circ = -1$

(5) $\cos(-225^\circ) = \cos 225^\circ = -\dfrac{\sqrt{2}}{2}$

(6) $\tan(-480^\circ) = \tan 240^\circ = \sqrt{3}$

問題 288

次の角を弧度法で表せ．

解答:

(1) $135^\circ = \dfrac{3\pi}{4}$　(2) $36^\circ = \dfrac{\pi}{5}$　(3) $-10^\circ = -\dfrac{\pi}{18}$

(4) $240^\circ = \dfrac{4\pi}{3}$　(5) $-190^\circ = -\dfrac{19\pi}{18}$

問題 289

次の角を 60 分法で表せ．

解答:

(1) $\dfrac{\pi}{3} = 60^\circ$　(2) $\dfrac{5\pi}{4} = 225^\circ$　(3) $-\dfrac{2\pi}{5} = -72^\circ$

(4) $\dfrac{7\pi}{3} = 420^\circ$　(5) $-\dfrac{\pi}{9} = -20^\circ$

問題 290

次の三角関数の値を求めよ．

解答:

(1) $\sin\dfrac{5\pi}{3} = -\sin\dfrac{\pi}{3} = -\dfrac{\sqrt{3}}{2}$

(2) $\cos\dfrac{5\pi}{4} = -\cos\dfrac{\pi}{4} = -\dfrac{\sqrt{2}}{2}$

(3) $\tan\dfrac{\pi}{6} = \dfrac{\sqrt{3}}{3}$

問題 291

弧の長さと扇形の面積を求めよ．

解答:

$l = r\theta$，$S = \dfrac{1}{2}r^2\theta$

(1) 半径 4，中心角 $\dfrac{\pi}{6}$：$l = \dfrac{2\pi}{3}$，$S = \dfrac{4\pi}{3}$

(2) 半径 3，弧 2π：$\theta = \dfrac{2\pi}{3}$，$S = 3\pi$

問題 292

次の等式を証明せよ．

解答:

(1) $\tan\theta + \dfrac{1}{\tan\theta} = \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \dfrac{1}{\sin\theta\cos\theta}$ □

(2) $\dfrac{1}{1-\cos\theta} + \dfrac{1}{1+\cos\theta} = \dfrac{2}{1-\cos^2\theta} = \dfrac{2}{\sin^2\theta}$ □

問題 293

象限と 1 つの三角関数の値から残りを求めよ．

解答:

(1) 第 3 象限，$\sin\theta = -\dfrac{4}{5}$

$\cos\theta = -\dfrac{3}{5}$，$\tan\theta = \dfrac{4}{3}$

(2) 第 4 象限，$\cos\theta = \dfrac{1}{3}$

$\sin\theta = -\dfrac{2\sqrt{2}}{3}$，$\tan\theta = -2\sqrt{2}$

(3) 第 3 象限，$\tan\theta = 3$

$\cos\theta = -\dfrac{\sqrt{10}}{10}$，$\sin\theta = -\dfrac{3\sqrt{10}}{10}$

問題 294

次の式を簡単にせよ．

解答:

(1) $\cos\theta\sin\left(\dfrac{\pi}{2}-\theta\right) + \sin\theta\cos\left(\dfrac{\pi}{2}-\theta\right) = \cos^2\theta + \sin^2\theta = 1$

(2) $\sin\left(\dfrac{\pi}{2}-\theta\right) - \sin(\pi+\theta) + \cos\left(\dfrac{\pi}{2}+\theta\right) + \cos(\pi-\theta)$
$= \cos\theta + \sin\theta - \sin\theta - \cos\theta = 0$

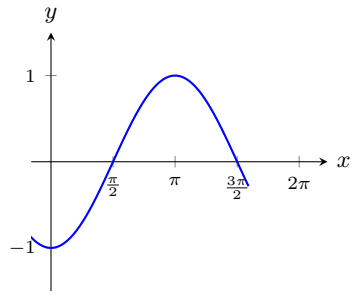
(3) $\tan(\pi+\theta)\sin\left(\dfrac{\pi}{2}+\theta\right) + \cos(\pi-\theta)\tan(\pi-\theta)$
$= \tan\theta\cos\theta + (-\cos\theta)(-\tan\theta) = \sin\theta + \sin\theta = 2\sin\theta$

問題 295

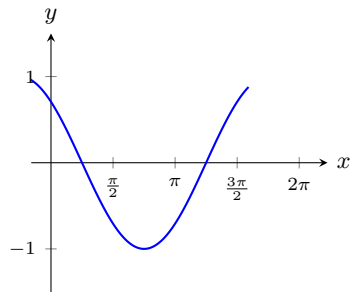
次の関数の周期を求め，グラフをかけ．

解答:

(1) $y = \sin\left(x - \dfrac{\pi}{2}\right) = -\cos x$　周期 2π



(2) $y = \cos\left(x + \dfrac{\pi}{4}\right)$　周期 2π

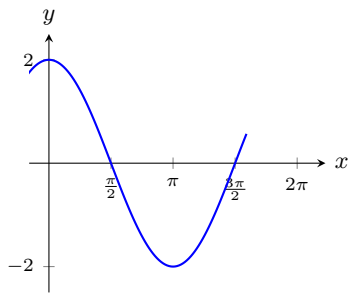


問題 296

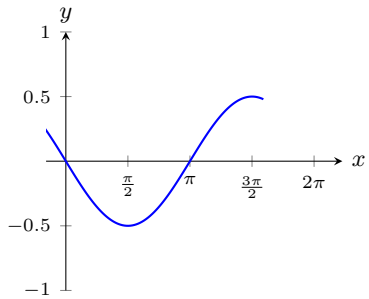
次の関数の周期を求め，グラフをかけ．

解答:

(1) $y = 2\cos x$　周期 2π，振幅 2



(2) $y = -\dfrac{1}{2}\sin x$　周期 2π，振幅 $\dfrac{1}{2}$



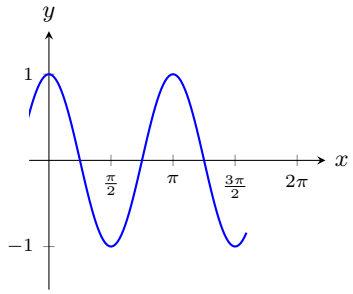
問題 297

次の関数の周期を求め，グラフをかけ．

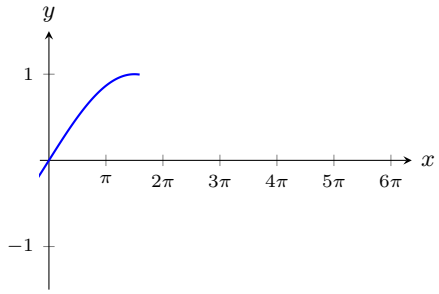
解答:

$y = \sin(kx)$ の周期は $\dfrac{2\pi}{|k|}$.

(1) $y = \cos 2x$　周期 π



(2) $y = \sin\dfrac{x}{3}$　周期 6π



問題 298

$0 \leqq x < 2\pi$ で方程式・不等式を解け．

解答:

(1) $\sin x = \dfrac{\sqrt{2}}{2} \to x = \dfrac{\pi}{4},\ \dfrac{3\pi}{4}$

(2) $\cos x = \dfrac{1}{2} \to x = \dfrac{\pi}{3},\ \dfrac{5\pi}{3}$

(3) $\sin x \geqq \dfrac{\sqrt{3}}{2} \to \dfrac{\pi}{3} \leqq x \leqq \dfrac{2\pi}{3}$

(4) $\cos x < -\dfrac{\sqrt{2}}{2} \to \dfrac{3\pi}{4} < x < \dfrac{5\pi}{4}$

問題 299

$0 \leqq x < 2\pi$ で方程式を解け．

解答:

(1) $\tan x = 0 \to x = 0,\ \pi$

(2) $\tan x = -1 \to x = \dfrac{3\pi}{4},\ \dfrac{7\pi}{4}$

問題 300

次の角を弧度法で表せ．

解答:

(1) $40^\circ = \dfrac{2\pi}{9}$　(2) $50^\circ = \dfrac{5\pi}{18}$　(3) $-18^\circ = -\dfrac{\pi}{10}$　(4) $-210^\circ = -\dfrac{7\pi}{6}$

問題 301

次の角を 60 分法で表せ．

解答:

(1) $\dfrac{7\pi}{6} = 210^\circ$　(2) $\dfrac{5\pi}{3} = 300^\circ$　(3) $-\dfrac{4\pi}{9} = -80^\circ$　(4) $-\dfrac{11\pi}{4} = -495^\circ$

問題 302

半径 6，中心角 $\dfrac{5\pi}{6}$ の扇形の弧の長さ l と面積 S を求めよ．

解答:

$l = 6 \cdot \dfrac{5\pi}{6} = 5\pi,\quad S = \dfrac{1}{2} \cdot 36 \cdot \dfrac{5\pi}{6} = 15\pi$

問題 303

次の三角関数の値を求めよ．

解答:

(1) $\sin\dfrac{11\pi}{6} = \sin\left(2\pi - \dfrac{\pi}{6}\right) = -\sin\dfrac{\pi}{6} = -\dfrac{1}{2}$

(2) $\cos\left(-\dfrac{5\pi}{4}\right) = \cos\dfrac{5\pi}{4} = -\cos\dfrac{\pi}{4} = -\dfrac{\sqrt{2}}{2}$

(3) $\tan\left(-\dfrac{5\pi}{3}\right) = \tan\left(-\dfrac{5\pi}{3} + 2\pi\right) = \tan\dfrac{\pi}{3} = \sqrt{3}$

問題 304

θ が第 2 象限で $\sin\theta = \dfrac{2}{3}$ のとき，$\cos\theta$，$\tan\theta$ を求めよ．

解答:

$\cos^2\theta = 1 - \dfrac{4}{9} = \dfrac{5}{9}$. 第 2 象限で $\cos < 0$：$\cos\theta = -\dfrac{\sqrt{5}}{3}$

$\tan\theta = \dfrac{2/3}{-\sqrt{5}/3} = -\dfrac{2}{\sqrt{5}} = -\dfrac{2\sqrt{5}}{5}$

問題 305

次の式を簡単にせよ．

解答:

(1) $\sin(\pi+\theta)\cos(2\pi-\theta) + \sin\left(\dfrac{\pi}{2}+\theta\right)\cos(\pi-\theta)$

$= (-\sin\theta)(\cos\theta) + (\cos\theta)(-\cos\theta) = -\sin\theta\cos\theta - \cos^2\theta$

$= -\cos\theta(\sin\theta + \cos\theta)$

(2) $\dfrac{\sin(\pi-\theta)}{\cos\left(\frac{\pi}{2}+\theta\right)} - \dfrac{\tan(\pi+\theta)\cos(\pi+\theta)}{\sin\left(\frac{3\pi}{2}-\theta\right)}$

$= \dfrac{\sin\theta}{-\sin\theta} - \dfrac{\tan\theta\cdot(-\cos\theta)}{-\cos\theta} = -1 - \tan\theta$

問題 306

次の等式を証明せよ．

解答:

(1) $\dfrac{1}{1+\sin\theta} + \dfrac{1}{1-\sin\theta} = \dfrac{2}{\cos^2\theta}$

左辺 $= \dfrac{(1-\sin\theta)+(1+\sin\theta)}{1-\sin^2\theta} = \dfrac{2}{\cos^2\theta} =$ 右辺　□

(2) $\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$

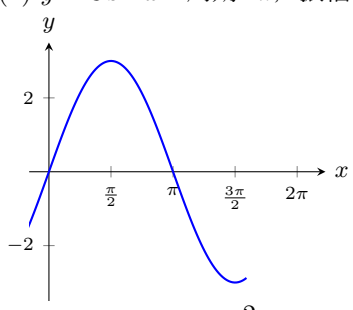
左辺 $= (\sin^2\theta + \cos^2\theta)(\sin^2\theta - \cos^2\theta) = 1\cdot(\sin^2\theta - \cos^2\theta) =$ 右辺　□

問題 307

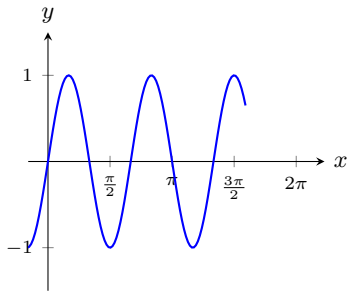
次の関数の周期とグラフをかけ．

解答:

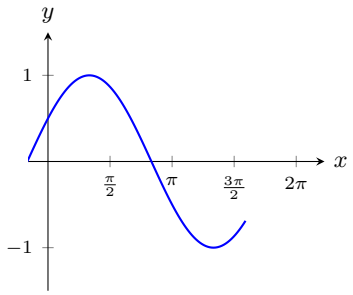
(1) $y = 3\sin x$　周期 2π，振幅 3



(2) $y = \sin 3x$　周期 $\dfrac{2\pi}{3}$



(3) $y = \cos\left(x - \dfrac{\pi}{3}\right)$　周期 2π



問題 308

$0 \leqq x < 2\pi$ で次を解け．

解答:

(1) $2\cos x + \sqrt{3} = 0 \to \cos x = -\dfrac{\sqrt{3}}{2}$

$x = \dfrac{5\pi}{6},\ \dfrac{7\pi}{6}$

(2) $\tan x + 1 \geqq 0 \to \tan x \geqq -1$

$\tan x = -1$ となるのは $x = \dfrac{3\pi}{4},\ \dfrac{7\pi}{4}$

$\therefore 0 \leqq x < \dfrac{\pi}{2}$ または $\dfrac{3\pi}{4} \leqq x < \dfrac{3\pi}{2}$ または $\dfrac{7\pi}{4} \leqq x < 2\pi$

（tan の定義域に注意：$x \neq \dfrac{\pi}{2},\ \dfrac{3\pi}{2}$）

問題 318

次の値を求めよ．

解答:

加法定理：$\sin(\alpha\pm\beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$, $\cos(\alpha\pm\beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$

(1) $\sin 75^\circ = \sin(45^\circ + 30^\circ) = \dfrac{\sqrt{2}}{2}\cdot\dfrac{\sqrt{3}}{2} + \dfrac{\sqrt{2}}{2}\cdot\dfrac{1}{2} = \dfrac{\sqrt{6}+\sqrt{2}}{4}$

(2) $\cos 15^\circ = \cos(45^\circ - 30^\circ) = \dfrac{\sqrt{2}}{2}\cdot\dfrac{\sqrt{3}}{2} + \dfrac{\sqrt{2}}{2}\cdot\dfrac{1}{2} = \dfrac{\sqrt{6}+\sqrt{2}}{4}$

(3) $\tan\dfrac{5\pi}{12} = \tan(75^\circ) = \tan(45^\circ + 30^\circ) = \dfrac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}} = \dfrac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3}$

問題 319

α が第 2 象限で $\sin\alpha = \dfrac{4}{5}$，$\beta$ が第 1 象限で $\cos\beta = \dfrac{5}{13}$ のとき，$\sin(\alpha+\beta)$，$\cos(\alpha-\beta)$ を求めよ．

解答:

$\cos\alpha = -\dfrac{3}{5}$（第 2 象限），$\sin\beta = \dfrac{12}{13}$（第 1 象限）

$\sin(\alpha+\beta) = \dfrac{4}{5}\cdot\dfrac{5}{13} + \left(-\dfrac{3}{5}\right)\cdot\dfrac{12}{13} = \dfrac{20}{65} - \dfrac{36}{65} = -\dfrac{16}{65}$

$\cos(\alpha-\beta) = \left(-\dfrac{3}{5}\right)\cdot\dfrac{5}{13} + \dfrac{4}{5}\cdot\dfrac{12}{13} = -\dfrac{15}{65} + \dfrac{48}{65} = \dfrac{33}{65}$

問題 320

$\sin\alpha = \dfrac{1}{\sqrt{5}}$, $\sin\beta = \dfrac{1}{\sqrt{10}}$（$\alpha, \beta$ は鋭角）のとき，$\alpha+\beta$ を求めよ．

解答:

$\cos\alpha = \dfrac{2}{\sqrt{5}}$，$\cos\beta = \dfrac{3}{\sqrt{10}}$

$\sin(\alpha+\beta) = \dfrac{1}{\sqrt{5}}\cdot\dfrac{3}{\sqrt{10}} + \dfrac{2}{\sqrt{5}}\cdot\dfrac{1}{\sqrt{10}} = \dfrac{3}{\sqrt{50}} + \dfrac{2}{\sqrt{50}} = \dfrac{5}{5\sqrt{2}} = \dfrac{1}{\sqrt{2}}$

$\alpha+\beta$ は鋭角の和なので $0 < \alpha+\beta < \pi$．$\sin(\alpha+\beta) = \dfrac{\sqrt{2}}{2}$ より

$\alpha+\beta = \dfrac{\pi}{4}$ または $\dfrac{3\pi}{4}$

$\cos(\alpha+\beta) = \dfrac{2}{\sqrt{5}}\cdot\dfrac{3}{\sqrt{10}} - \dfrac{1}{\sqrt{5}}\cdot\dfrac{1}{\sqrt{10}} = \dfrac{5}{\sqrt{50}} = \dfrac{1}{\sqrt{2}} > 0$ より

$\alpha+\beta = \dfrac{\pi}{4}$

問題 321

2 倍角の公式を用いて，次の値を求めよ．

解答:

$\sin 2\alpha = 2\sin\alpha\cos\alpha$，$\cos 2\alpha = 1 - 2\sin^2\alpha = 2\cos^2\alpha - 1$

(1) $\sin\dfrac{\pi}{8}$

$\cos\dfrac{\pi}{4} = 1 - 2\sin^2\dfrac{\pi}{8}$，$\dfrac{\sqrt{2}}{2} = 1 - 2\sin^2\dfrac{\pi}{8}$

$\sin^2\dfrac{\pi}{8} = \dfrac{2-\sqrt{2}}{4}$，$\sin\dfrac{\pi}{8} = \dfrac{\sqrt{2-\sqrt{2}}}{2}$

(2) $\cos\dfrac{\pi}{8}$

$\cos\dfrac{\pi}{4} = 2\cos^2\dfrac{\pi}{8} - 1$，$\cos^2\dfrac{\pi}{8} = \dfrac{2+\sqrt{2}}{4}$

$\cos\dfrac{\pi}{8} = \dfrac{\sqrt{2+\sqrt{2}}}{2}$

問題 322

θ が第 3 象限で $\cos\theta = -\dfrac{3}{4}$ のとき，$\sin 2\theta$，$\cos 2\theta$，$\tan 2\theta$ を求めよ．

解答:

$\sin^2\theta = 1 - \dfrac{9}{16} = \dfrac{7}{16}$．第 3 象限で $\sin < 0$：$\sin\theta = -\dfrac{\sqrt{7}}{4}$

$\sin 2\theta = 2\cdot\left(-\dfrac{\sqrt{7}}{4}\right)\left(-\dfrac{3}{4}\right) = \dfrac{3\sqrt{7}}{8}$

$\cos 2\theta = 2\cos^2\theta - 1 = 2\cdot\dfrac{9}{16} - 1 = \dfrac{1}{8}$

$\tan 2\theta = \dfrac{\sin 2\theta}{\cos 2\theta} = \dfrac{3\sqrt{7}/8}{1/8} = 3\sqrt{7}$

問題 323

三角関数の合成を行え．

解答:

$a\sin\theta + b\cos\theta = \sqrt{a^2+b^2}\sin(\theta+\varphi)$（$\tan\varphi = \dfrac{b}{a}$）

(1) $\sin\theta + \cos\theta = \sqrt{2}\sin\left(\theta + \dfrac{\pi}{4}\right)$

(2) $\sin\theta - \cos\theta = \sqrt{2}\sin\left(\theta - \dfrac{\pi}{4}\right)$

(3) $\sqrt{3}\sin\theta + \cos\theta = 2\sin\left(\theta + \dfrac{\pi}{6}\right)$

(4) $\sin\theta - \sqrt{3}\cos\theta = 2\sin\left(\theta - \dfrac{\pi}{3}\right)$

問題 324

$0 \leqq \theta < 2\pi$ のとき，$y = \sin\theta + \cos\theta$ の最大値と最小値を求めよ．

解答:

$y = \sqrt{2}\sin\left(\theta + \dfrac{\pi}{4}\right)$

$-1 \leqq \sin\left(\theta + \dfrac{\pi}{4}\right) \leqq 1$ より

最大値 $\sqrt{2}$（$\theta = \dfrac{\pi}{4}$），最小値 $-\sqrt{2}$（$\theta = \dfrac{5\pi}{4}$）

問題 325

$0 \leqq \theta < 2\pi$ のとき，$\sqrt{3}\sin\theta - \cos\theta = 1$ を解け．

解答:

$\sqrt{3}\sin\theta - \cos\theta = 2\sin\left(\theta - \dfrac{\pi}{6}\right) = 1$

$\sin\left(\theta - \dfrac{\pi}{6}\right) = \dfrac{1}{2}$

$\theta - \dfrac{\pi}{6} = \dfrac{\pi}{6}$，$\dfrac{5\pi}{6}$

$\theta = \dfrac{\pi}{3}$，π

問題 326

次の値を求めよ．

解答:

(1) $\cos\dfrac{5\pi}{12} = \cos(75^\circ) = \cos(45^\circ + 30^\circ) = \dfrac{\sqrt{6}-\sqrt{2}}{4}$

(2) $\sin\dfrac{\pi}{12} = \sin(15^\circ) = \sin(45^\circ - 30^\circ) = \dfrac{\sqrt{6}-\sqrt{2}}{4}$

(3) $\tan\dfrac{7\pi}{12} = \tan(105^\circ) = \tan(60^\circ{+}45^\circ) = \dfrac{\sqrt{3}+1}{1-\sqrt{3}} = -(2+\sqrt{3})$

問題 327

$\sin\alpha = \dfrac{3}{5}$（$\alpha$ 鋭角），$\cos\beta = -\dfrac{12}{13}$（$\beta$ 鈍角）のとき，$\cos(\alpha+\beta)$ を求めよ．

解答:

$\cos\alpha = \dfrac{4}{5}$，$\sin\beta = \dfrac{5}{13}$

$\cos(\alpha{+}\beta) = \dfrac{4}{5}\cdot\left(-\dfrac{12}{13}\right) - \dfrac{3}{5}\cdot\dfrac{5}{13} = -\dfrac{48}{65} - \dfrac{15}{65} = -\dfrac{63}{65}$

問題 328

θ が第 1 象限で $\sin\theta = \dfrac{5}{13}$ のとき，$\sin 2\theta$，$\cos 2\theta$ を求めよ．

解答:

$\cos\theta = \dfrac{12}{13}$

$\sin 2\theta = 2\cdot\dfrac{5}{13}\cdot\dfrac{12}{13} = \dfrac{120}{169}$

$\cos 2\theta = 1 - 2\cdot\dfrac{25}{169} = \dfrac{119}{169}$

問題 329

$0 \leqq \theta < 2\pi$ で $\sin\theta + \sqrt{3}\cos\theta = -1$ を解け．

解答:

$\sin\theta + \sqrt{3}\cos\theta = 2\sin\left(\theta + \dfrac{\pi}{3}\right) = -1$

$\sin\left(\theta + \dfrac{\pi}{3}\right) = -\dfrac{1}{2}$

$\theta + \dfrac{\pi}{3} = \dfrac{7\pi}{6},\ \dfrac{11\pi}{6}$

$\theta = \dfrac{5\pi}{6},\ \dfrac{3\pi}{2}$

問題 330

$0 \leqq \theta < 2\pi$ のとき，$y = \sin\theta - \sqrt{3}\cos\theta$ の最大値と最小値を求めよ．

解答:

$y = 2\sin\left(\theta - \dfrac{\pi}{3}\right)$

最大値 2 $(\theta - \dfrac{\pi}{3} = \dfrac{\pi}{2},\ \theta = \dfrac{5\pi}{6})$，最小値 -2 $(\theta = \dfrac{11\pi}{6})$

問題 331

$\tan\alpha + \tan\beta = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta}$ を証明せよ．

解答:

左辺 $= \dfrac{\sin\alpha}{\cos\alpha} + \dfrac{\sin\beta}{\cos\beta} = \dfrac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta} = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta} =$ 右辺 □

問題 332

$\sin(\alpha+\beta)\sin(\alpha-\beta) = \sin^2\alpha - \sin^2\beta$ を証明せよ．

解答:

左辺 $= (\sin\alpha\cos\beta + \cos\alpha\sin\beta)(\sin\alpha\cos\beta - \cos\alpha\sin\beta)$

$= \sin^2\alpha\cos^2\beta - \cos^2\alpha\sin^2\beta$

$= \sin^2\alpha(1-\sin^2\beta) - (1-\sin^2\alpha)\sin^2\beta$

$= \sin^2\alpha - \sin^2\alpha\sin^2\beta - \sin^2\beta + \sin^2\alpha\sin^2\beta = \sin^2\alpha - \sin^2\beta =$ 右辺 □

問題 333

2 直線 $y = \dfrac{1}{2}x + 1$，$y = 2x - 3$ のなす角 θ を求めよ．

解答:

傾き $m_1 = \dfrac{1}{2}$，$m_2 = 2$ とすると

$\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\dfrac{\frac{1}{2} - 2}{1+1}\right| = \left|\dfrac{-\frac{3}{2}}{2}\right| = \dfrac{3}{4}$

$\theta = \arctan\dfrac{3}{4} \approx 36.87^\circ$

問題 334

$\sin 2\theta = 2\sin\theta\cos\theta$ と $\cos 2\theta = \cos^2\theta - \sin^2\theta$ を用いて $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$ を導け．

解答:

$\tan 2\theta = \dfrac{\sin 2\theta}{\cos 2\theta} = \dfrac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$

分子分母を $\cos^2\theta$ で割る：

$= \dfrac{2\frac{\sin\theta}{\cos\theta}}{1 - \frac{\sin^2\theta}{\cos^2\theta}} = \dfrac{2\tan\theta}{1-\tan^2\theta}$ □

問題 335

$\cos^4\theta - \sin^4\theta = \cos 2\theta$ を示せ．

解答:

左辺 $= (\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta) = 1\cdot\cos 2\theta =$ 右辺 □

問題 336

$0 \leqq \theta < 2\pi$ のとき $2\cos^2\theta + \sin\theta - 1 = 0$ を解け.

解答:

$\cos^2\theta = 1 - \sin^2\theta$ を代入：

$2(1-\sin^2\theta) + \sin\theta - 1 = 0,\ -2\sin^2\theta + \sin\theta + 1 = 0$

$2\sin^2\theta - \sin\theta - 1 = 0,\ (2\sin\theta + 1)(\sin\theta - 1) = 0$

$\sin\theta = -\dfrac{1}{2}$ または $\sin\theta = 1$

$\sin\theta = 1$：$\theta = \dfrac{\pi}{2}$

$\sin\theta = -\dfrac{1}{2}$：$\theta = \dfrac{7\pi}{6},\ \dfrac{11\pi}{6}$

$\therefore \theta = \dfrac{\pi}{2},\ \dfrac{7\pi}{6},\ \dfrac{11\pi}{6}$